Scores and Calibration Intuition

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Responsible Plant Classifier

THE CONNECTED WEEK

One readable group-probability evidence record.

Trace mean forecasts, positive fractions and the limits of calibration evidence.

Keep one habit: W1–W3/W5 → selected two-group task → project score record → changed-case return.

MAIN LESSON · SECTION 1

Recall · Retrieve before notes

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Train a Classifier. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Decision Trees. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Leakage. Give an example and a mistake that the idea helps you avoid.

Train a Classifier

> **My remembered idea, example and correction**

Decision Trees

> **My remembered idea, example and correction**

Leakage

> **My remembered idea, example and correction**

W1 · Explain the mechanism

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Main score means p(positive):10cases at0.2with2positives→point(0.2,0.2);10at0.8with6positives→(0.8,0.6),sample overstatement0.2. Separate picture has10at0.8with8positives and10at0.2with4positives; low group understates by0.2. Changing pictured low positives to2matches0.2there. Main30-card lab adds a middle group10at0.5with7positives; points low(0.2,0.2),middle(0.5,0.7),high(0.8,0.6).

  Chosen-label correctness is a different event. Small groups, unobserved scores and changed populations limit conclusions.

- Reference for this exercise: open this week's student lesson, "Scores and Calibration Intuition", at "See how it works" and "Follow a worked example". Use the original worked values or code named in the question; the later pictorial case is a separate example. Keep this reference beside your workbook while practising.

- Give the observed frequencies for the two worked groups.

> **My reasoning and evidence**

W2 · Trace the current experiment

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Main score means p(positive):10cases at0.2with2positives→point(0.2,0.2);10at0.8with6positives→(0.8,0.6),sample overstatement0.2. Separate picture has10at0.8with8positives and10at0.2with4positives; low group understates by0.2. Changing pictured low positives to2matches0.2there. Main30-card lab adds a middle group10at0.5with7positives; points low(0.2,0.2),middle(0.5,0.7),high(0.8,0.6).

  Chosen-label correctness is a different event. Small groups, unobserved scores and changed populations limit conclusions.

- Reference for this exercise: open this week's student lesson, "Scores and Calibration Intuition", at "See how it works" and "Follow a worked example". Use the original worked values or code named in the question; the later pictorial case is a separate example. Keep this reference beside your workbook while practising.

- Which worked group overstates positive frequency and by how much?

> **My reasoning and evidence**

W3 · Repair the reasoning

> Keep this task’s complete inputs, stated rule and support record together.

Task and evidence record

- Main score means p(positive):10cases at0.2with2positives→point(0.2,0.2);10at0.8with6positives→(0.8,0.6),sample overstatement0.2. Separate picture has10at0.8with8positives and10at0.2with4positives; low group understates by0.2. Changing pictured low positives to2matches0.2there. Main30-card lab adds a middle group10at0.5with7positives; points low(0.2,0.2),middle(0.5,0.7),high(0.8,0.6).

  Chosen-label correctness is a different event. Small groups, unobserved scores and changed populations limit conclusions.

- Why can constant .5 **prediction**s look calibrated but rank poorly?

> **My reasoning and evidence**

MAIN LESSON · SECTION 16

W5 · Plan the next check

> Keep this task’s complete inputs, stated rule and support record together.

Task and evidence record

- Main score means p(positive):10cases at0.2with2positives→point(0.2,0.2);10at0.8with6positives→(0.8,0.6),sample overstatement0.2. Separate picture has10at0.8with8positives and10at0.2with4positives; low group understates by0.2. Changing pictured low positives to2matches0.2there. Main30-card lab adds a middle group10at0.5with7positives; points low(0.2,0.2),middle(0.5,0.7),high(0.8,0.6).

  Chosen-label correctness is a different event. Small groups, unobserved scores and changed populations limit conclusions.

- What is wrong with calibrating and reporting success on the same final-test records?

> **My reasoning and evidence**

MAIN LESSON · SECTION 4

Try alone · Selected experiment task

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- In this selected task the scores mean p(positive). Positive outcomes are the supplied counts, not counts of correct chosen labels. Plot group mean probability on x and observed positive frequency on y; show the two denominators and state the limited evidence. No per-case prediction decision rule, global calibration test or future guarantee is supplied.

- Twenty cases all predict .3 with eight positives; ten all predict .7 with seven positives. Give plot points and interpret.

- Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Independent case: Apply independently

> **My independent explanation, trace and limits**
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MAIN LESSON · SECTION 9

Project · Keep the experiment evidence

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Add an uncertainty band to the Plant Classifier and define what the interface says when **score**s fall inside it.

- The project change and **evidence** I will keep

Write or draw your reasoning and evidence.

MAIN LESSON · SECTION 17

Feedback · Repair and return

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

MAIN LESSON · SECTION 10

Finish · Judge your evidence

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- - I can explain the mechanism.
  - I can show evidence.
  - I can handle a changed case.
  - I can name a limit.

  Mark each as not yet, with help or independently, and attach supporting work.

- Teacher review _____ Next step _____

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Use W2: Locate the discrepancy for **Calibration** as a group property. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

▼ Extra practice · Individual reasoning check

REFERENCE · SECTION 3

Extra practice · Individual reasoning check

> Keep this task’s complete inputs, stated rule and support record together.

Task and evidence record

- Use the worked case for **Calibration** as a group property. Proposed explanation: “A final answer is enough”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Earlier learning target

REFERENCE · SECTION 5

Reference · Earlier learning target

> Keep this task’s complete inputs, stated rule and support record together.

Task and evidence record

- Twenty cases receive **score** .7 and fourteen are correct. Explain what the observed rate suggests and what it does not establish about another **score** group.

▼ **Extra practice · Earlier complete case**

REFERENCE · SECTION 6

Extra practice · Earlier complete case

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Ten cases receive probability .8 for spotted; eight are spotted.
- Interpret **calibration** and its limits.
- Show your trace, calculation or **evidence**

Write or draw your reasoning and evidence.

▼ Extra practice · Changed case

REFERENCE · SECTION 7

Extra practice · Changed case

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Among 20 **prediction**s with **score** 0.8, 12 are correct. What is observed accuracy in this group and how does it compare with 0.8?
- My independent answer and reason

> **Write or draw your reasoning and evidence.**

Week 14 Test and reflect

▼ Extra practice · Design a revealing test

REFERENCE · SECTION 8

Extra practice · Design a revealing test

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

▼ **Extra practice · Earlier transfer**

REFERENCE · SECTION 11

Extra practice · Earlier transfer

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Twenty cases receive **score** .7 and fourteen are correct. Explain what the observed rate suggests and what it does not establish about another **score** group.

- Record support used: none, hint or teacher explanation.

My independently reasoned answer

▼ **Reference · Earlier W4 record**

REFERENCE · SECTION 15

Reference · Earlier W4 record

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Twenty cases all predict .3 with eight positives; ten all predict .7 with seven positives. Give plot points and interpret.

My reasoning and evidence

▼ **Extra practice · Pictured case**

REFERENCE · SECTION 18

Extra practice · Pictured case

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Treat **calibration** as a comparison across groups of probability **score**s.
- Toy classifier **output**s p(spotted)=0.8 on ten cases; eight are actually spotted. Another ten have p=0.2; four are spotted.
- All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.
- Does a calibrated 0.8 **score** mean every such leaf is spotted? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.
- Support used: alone / hint / worked example. This record is practice **evidence**.
- Project connection: For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

  Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

AI ACADEMY / AI-4/14

Scores and Calibration Intuition

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.

1. Group scores

Keep the ten 0.8 cases and ten 0.2 cases separate.

↓

2. Compare frequencies

Observed spotted fractions are 8/10=0.8 and 4/10=0.4.

↓

3. Interpret gap

The 0.8 group matches here; the 0.2 group understates the observed fraction by 0.2.

CHANGE ONE THING

The 0.2 group has two spotted cases instead of four.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

My picture-based explanation and changed-case prediction

REFERENCE · SECTION 19

Reference · Project portfolio milestone

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Project: Responsible Plant Classifier
- Current milestone: Test changed cases. Keep normal, boundary and failure cases; compare expected and observed results.
- For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

  Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

▼ Extra practice · Predict, check and repair

REFERENCE · SECTION 20

Extra practice · Predict, check and repair

> Keep supported feedback separate from the reserved independent case.

First supported question and hint

- Treat **calibration** as a comparison across groups of probability **score**s.
- Which labelled step matches this detail? The 0.8 group matches here; the 0.2 group understates the observed fraction by 0.2.
- 1. Group **score**s
- 2. Compare frequencies
- 3. Interpret gap
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: Interpret gap The 0.8 group matches here; the 0.2 group understates the observed fraction by 0.2. Read the picture in order.

The 0.8 group matches here; the 0.2 group understates the observed fraction by 0.2. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Changed-case question and hint

- Changed-case practice: The 0.2 group has two spotted cases instead of four. Which conclusion is supported by the supplied facts?
- 1. Its observed fraction becomes 0.2 and matches the supplied **score** in this group.
- 2. This one example guarantees correct results for every future **input**.
- 3. We can decide the result without examining the changed **input** or the procedure.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the changed explanation**

After your attempt, compare with the supported explanation: Its observed fraction becomes 0.2 and matches the supplied **score** in this group. **Calibration** compares stated probabilities with group outcome frequencies.

Repair and keep the record

Review Train a Classifier (ai-4/13). Goal: Connect **training**-only scaling and neighbour **prediction** in one pipeline. Short practice: Which pipeline record would help another team reproduce the result?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Decision Trees (ai-4/11). Goal: Fit a root question and leaf **prediction**s from **training** examples. Short practice: Which **input** distinguishes the original and changed tree?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Leakage (ai-4/8). Goal: Choose a split unit that matches the intended **prediction** claim. Short practice: How many independently new plant identities are in the original test?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

> **What I repaired, the evidence I kept and the next test**
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▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Predict, run and debug

Keep your completed work

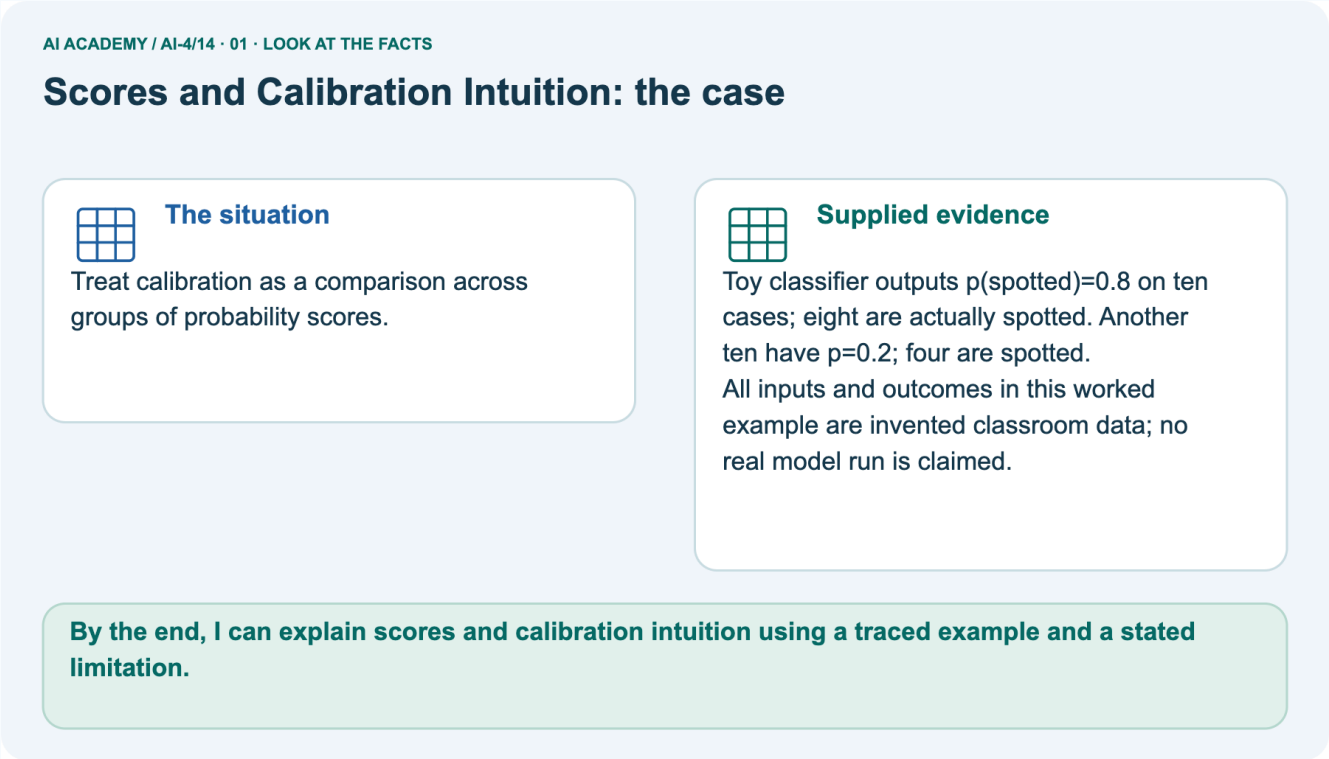
▼ **Reference · Six-picture response sheets**

REFERENCE · SECTION 21

Reference · Six-picture response sheets

> Choose the relevant picture instead of repeating every task.

01. Scores and Calibration Intuition: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

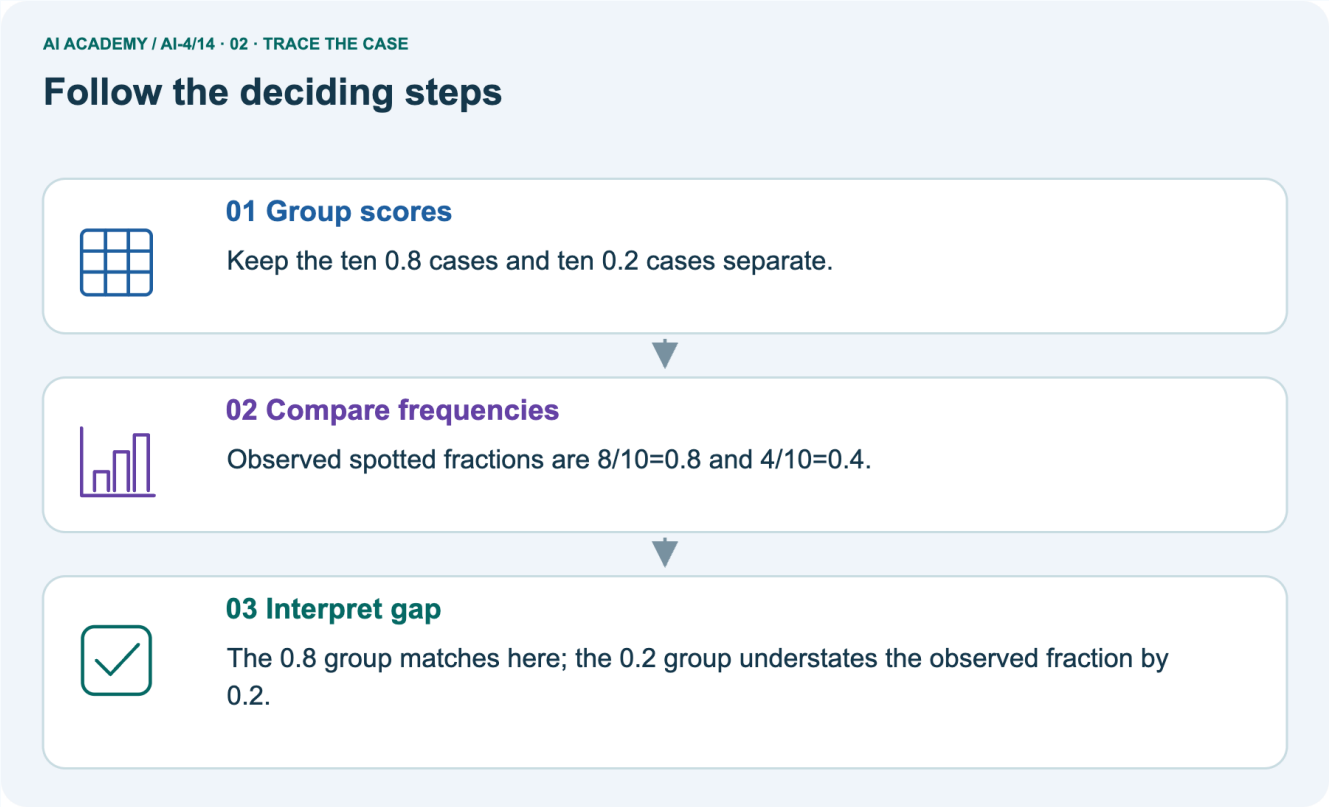
Treat **calibration** as a comparison across groups of probability **score**s. Supplied facts: Toy classifier **output**s p(spotted)=0.8 on ten cases; eight are actually spotted. Another ten have p=0.2; four are spotted.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Group **score**s: Keep the ten 0.8 cases and ten 0.2 cases separate. → Compare frequencies: Observed spotted fractions are 8/10=0.8 and 4/10=0.4. → Interpret gap: The 0.8 group matches here; the 0.2 group understates the observed fraction by 0.2.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Four-step worked evidence for Week 14: Scores and Calibration

Intuition

AI ACADEMY / AI-4/14 · 03 · INSPECT THE LESSON MODEL

Four-step worked evidence for Week 14: Scores and Calibration Intuition

Step	Evidence and reasoning
1	Each score estimates the probability that the selected label is correct. Ten predictions have such scores near 0.8.
2	Only six are correct, so observed correctness is 6 ÷ 10 = 60%.
3	The score ranking may still help, but 0.8 should not be stated as an 80% guarantee.
4	Use bins across many cases and design an uncertain zone requiring human review.

Trace these corrected steps. Keep the stated metric, assumptions and interpretation when you change the case.

Follow the labelled picture with the matching explanation. Open the vector diagram

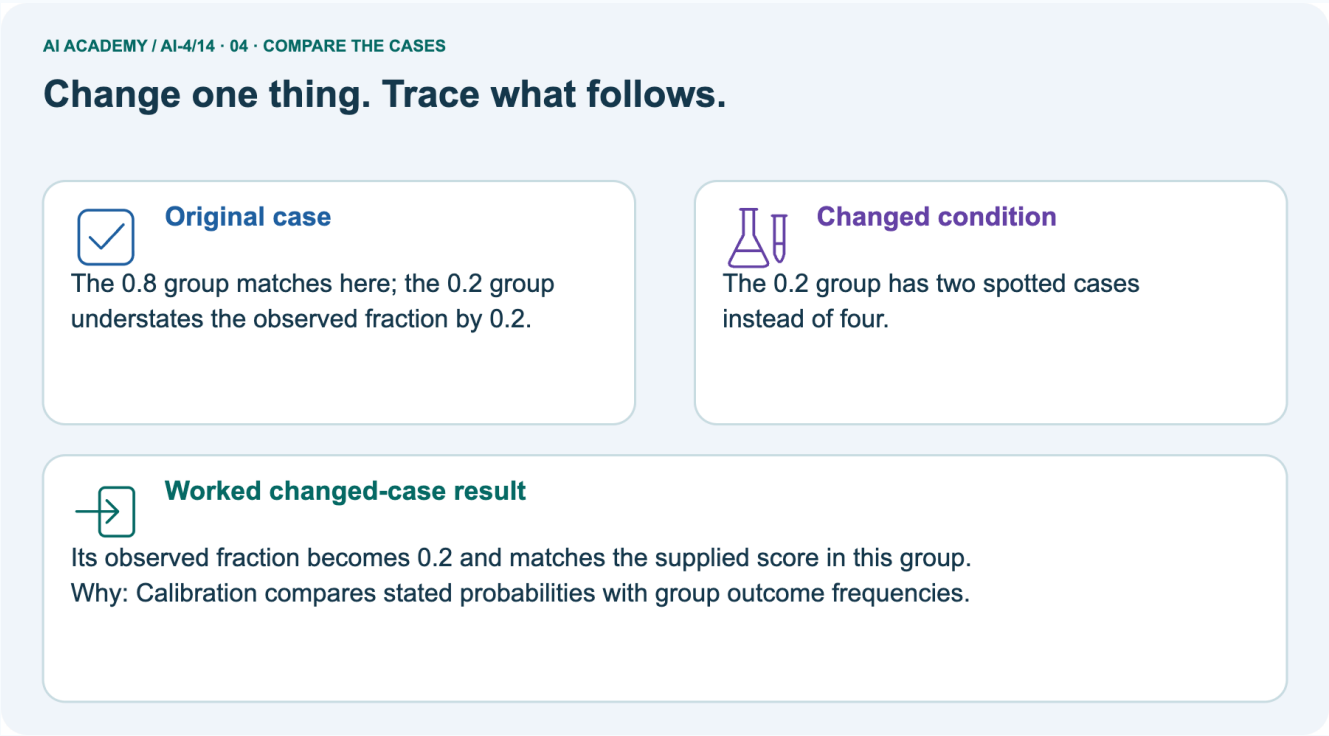
Four-step worked **evidence** for Week 14: **Score**s and **Calibration** Intuition. 1 | Each **score** estimates the probability that the selected label is correct.

Ten **prediction**s have such **score**s near 0.8.. 2 | Only six are correct, so observed correctness is 6 ÷ 10 = **60%**.. 3 | The **score** ranking may still help, but 0.8 should not be stated as an **80%** guarantee.. 4 | Use bins across many cases and design an uncertain zone requiring human review.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

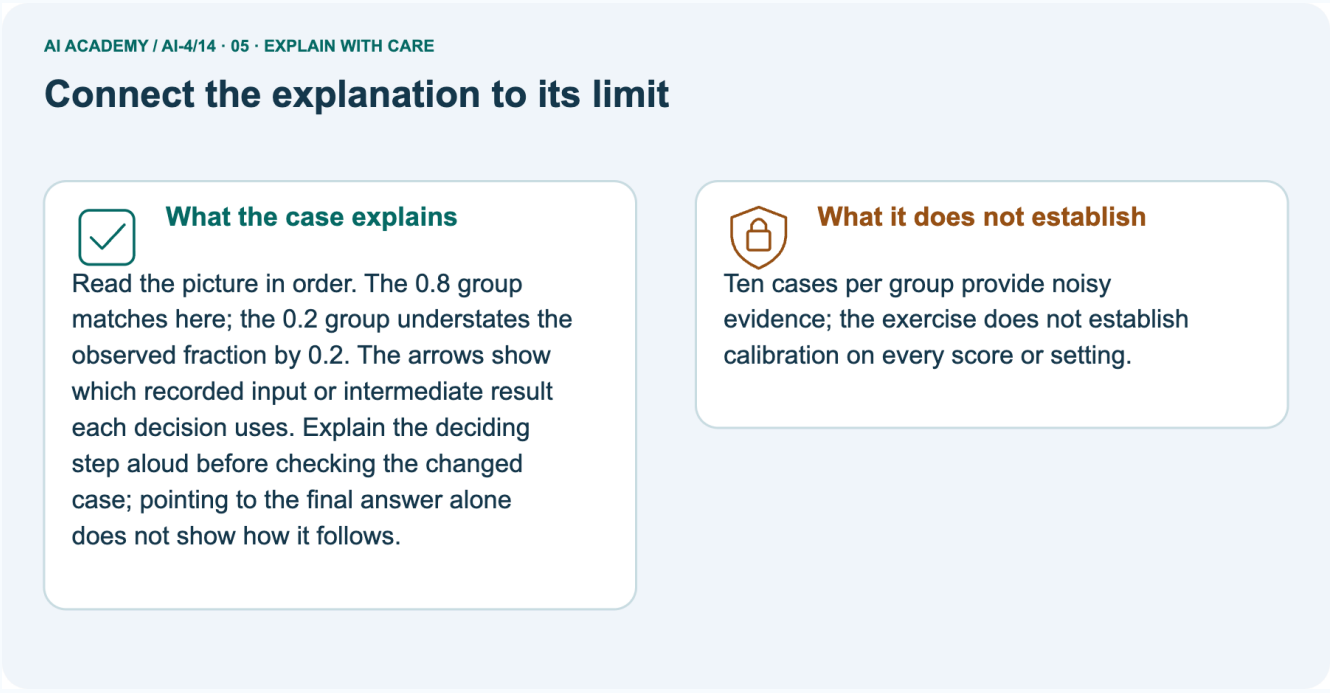
Original case: The 0.8 group matches here; the 0.2 group understates the observed fraction by 0.2. Changed condition: The 0.2 group has two spotted cases instead of four.

Expected result: Its observed fraction becomes 0.2 and matches the supplied **score** in this group. Reason: **Calibration** compares stated probabilities with group outcome frequencies.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture in order. The 0.8 group matches here; the 0.2 group understates the observed fraction by 0.2. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Limit: Ten cases per group provide noisy **evidence**; the exercise does not establish **calibration** on every **score** or setting.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Does a calibrated 0.8 **score** mean every such leaf is spotted? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

> **My picture explanation, deciding evidence and one limit**

▼ **Earlier wording, alternatives and picture transcripts**

01. Scores and Calibration Intuition: the case

02. Follow the deciding steps

03. Four-step worked evidence for Week 14: Scores and Calibration Intuition

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ **Optional interactive practice and completed work**

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.